QUANTITATIVE FINANCE / AI

Atharv Sharma

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EDUCATION

Cuthbertson High School

Waxhaw, NC — Current 9th Grade Student

TECHNICAL SKILLS

Python
AI / Machine Learning
Data Science
SQL
JavaScript / TypeScript
React / Next.js
Streamlit
Tableau / Power BI
APIs
Git / GitHub

INTERESTS

Quantitative Finance, Machine Learning, Software Engineering, Data Science,

SUMMARY

9th grade student focused on quantitative finance, machine learning, and data science. Sole author of a return-predictability paper currently in major revision at NHSJS, and builder of AI applications spanning LLM fine-tuning, financial analysis, and economic dashboards. Skilled in Python, SQL, and data visualization, with a track record of taking projects from research question to working product.

RESEARCH

Does Deviation from Prior Earnings-Call Language Predict Stock Returns? A TF-IDF Pilot Study on SEC EDGAR Transcripts

Sole author · Zenodo, Aug 2026 · DOI 10.5281/zenodo.21875292 · Major revision, NHSJS

- Tested whether year-over-year change in a firm's own earnings-call language is associated with subsequent returns, using TF-IDF cosine similarity on verbatim Exhibit 99 transcripts scraped from SEC EDGAR — the first such test on fully public data.
- Built the end-to-end pipeline across 24 firms and 358 firm-quarters (2018–2024): the long-short deviation portfolio earned 1.27%/month (t = 0.88) against Fama-French five factors plus momentum — economically large, statistically null.
- Power analysis showed the test could only detect effects ≥ 6%/month, so the null is a detection-limit finding, not a rejection. Revised after peer review to narrow the claim, add a sentiment benchmark, and expand to 30 verified references.

PROJECTS

Llama QLoRA Chatbot & Prompt-Injection Defense Harness

- Fine-tuned Llama 3.1 8B Instruct with QLoRA (rank-16 adapter) on OASST1 and Dolly-15k, sized to train end-to-end on a single 32GB consumer GPU.
- Built an evaluation harness testing the model against documented injection attacks, scored by an LLM judge.
- Added an instruction-hierarchy and input-classifier defense layer that cut attack success rate from 40% to 0% on the five-attack battery.

Math Scan AI

- AI-powered app that evaluates student math work using OCR and gives detailed feedback; awarded 2nd Place at a science fair.

Economic Data Dashboard

- Built an interactive dashboard visualizing GDP, inflation, unemployment, and other economic indicators.

Stock Analysis Platform

- Python application for stock performance analysis with historical price charts and automated financial calculations.

EXPERIENCE

Software Engineer, Moonshot (YC)

Aug 2026 – Present

Independent Software Developer

- Independently design, develop, and maintain software projects focused on AI, finance, and data science.
- Manage project timelines from research and planning through deployment and iteration.

LEADERSHIP

- Take projects from idea to working product independently, then present the result.
- Built **QuantifyAI** solo and won **1st Place, Education** at a Minorities in STEM hackathon.
- Contribute to **FinanceMeta**, developing financial-literacy project ideas and member engagement.

AWARDS

- 1st Place, Education — Minorities in STEM hackathon, for QuantifyAI
- 2nd Place — Science Fair, for Math Scan AI